

$$I = \frac{q}{t} = \frac{dq}{dt} = \frac{dQ}{dt} = C \frac{dV}{dt} \quad q = I dt$$

$$v(t) = \frac{1}{C} \int_{-\infty}^{t_0} i(s) ds + \frac{1}{C} \int_{t_0}^t i(s) ds \rightarrow \text{随时间变化的电流值}$$

$$= V(t_0) + \frac{1}{C} \int_{t_0}^t i(s) ds$$

$$q = I dt \quad q = \int I(s) ds = \int_{-\infty}^{t_0} I(s) ds + \int_{t_0}^t I(s) ds$$

$$= q(t_0) + \int_{t_0}^t I(s) ds$$

$$\int_1 p = v i = v C \frac{dV}{dt} \rightarrow \begin{array}{l} V+ \rightarrow V \uparrow + \rightarrow ++ \text{吸收} \\ \quad \quad \quad \searrow V \downarrow - \rightarrow +- \text{放电} \\ V- \rightarrow V \uparrow + \rightarrow -+ \text{放电} \\ \quad \quad \quad \searrow V \downarrow - \rightarrow -- \text{吸收} \end{array}$$

$$W_C = P t = \int P dt = \int v i = \int v C \frac{dV}{dt} dt$$

$$= \int_{-\infty}^t C V dV = \frac{1}{2} C V^2 = \frac{1}{2} C V^2(t) - \frac{1}{2} C V^2(-\infty)$$

$$= \frac{1}{2} C V^2(t)$$

$$W_C = \frac{1}{2} C V^2(t) - \frac{1}{2} C V^2(t_0)$$